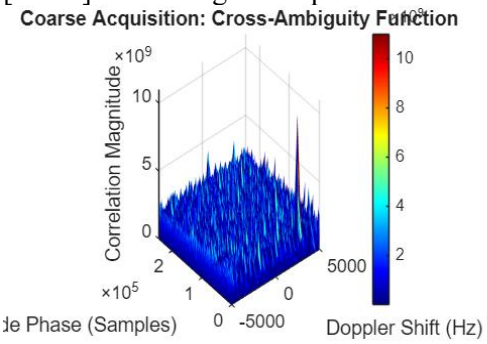
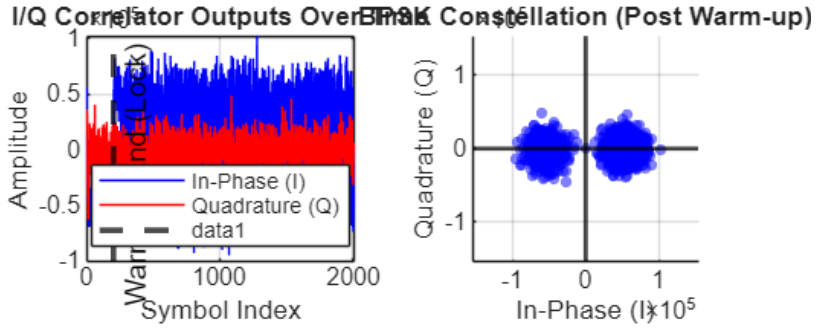
[Part 1] Signal Generation & Cryptographic Initialization
[Init] Generating ECDSA (secp256r1) Key Pair...
[Tx] Generating TESLA Hash Chain (n=10)...
[Tx] Deterministic puncturing and Payload Insertion.

[Part 2] Coarse Signal Acquisition



[Part 3] Signal Tracking (Realistic Coupled PLL & DLL)
[!] End of signal reached at symbol 2010



[Part 4] CNAV-2 Data Channel Decoding (BCH & LDPC)
[Rx] Subframe 1 TOI (BCH) Recovered: 0
[Rx] Subframe 2 (LDPC): CRC Passed!
[Rx] Subframe 3 (LDPC): CRC Passed!
[Rx] Receiver successfully reconstructed the pure raw subframes (926 bits).

[Part 5] Physical Payload Extraction & Viterbi Decoding
[Rx] Viterbi Decoding Errors: 0 / 900

[Part 6] Cryptographic Verification (Receiver Side)
[Rx] Pilot CRC Check : PASS
[Rx] TESLA Key Chain Verify : PASS (Hash matched stored commitment K_3)
[Rx] Root Signature Verify : PASS (ECDSA Validates Derived K_0)
[Rx] Navigation MAC Verify : PASS (Authentication Payload are Authentic & Intact)

[+++] Cryptographic Authentication Fully Successful! Signal is Genuine. [+++]